British Airways Passenger-CSAT Dashboard

Introduction

Real British Airways (BA) is one of the world's leading global airlines, serving millions of passengers every year across an extensive international network. In today's highly competitive airline industry, customer satisfaction has become a critical differentiator for retaining passengers, strengthening brand loyalty, and sustaining long-term profitability. With the rapid growth of digital platforms, airlines now receive large volumes of customer feedback through surveys, online reviews, and service ratings. This data, if analyzed effectively, provides valuable insights into passenger experiences, service quality, and operational performance.

The **British Airways Passenger-CSAT Dashboard project** aims to transform raw customer satisfaction and survey review data into meaningful, actionable insights using data analytics and visualization techniques. The dataset consists of passenger ratings collected across multiple dimensions such as overall satisfaction, seat comfort, cabin staff service, food and beverages, ground service, value for money, and entertainment. The feedback spans various countries, aircraft types, seat classes, and traveler categories, allowing for a comprehensive evaluation of service performance.

The primary objective of this project is to support airline management, flight operations teams, and business analysts by providing a **dynamic and interactive Power BI dashboard**. This dashboard enables users to explore customer satisfaction metrics in real time using filters and slicers such as country, aircraft type, seat class, traveler type, and time period. By leveraging these capabilities, decision-makers can quickly identify service gaps, monitor trends, and prioritize improvement initiatives.

Overall, this project demonstrates an end-to-end data analytics workflow, including data cleaning, transformation, modeling, DAX calculations, and advanced visualization, to deliver a user-friendly decision-support tool for enhancing passenger experience and operational excellence at British Airways.

KPI Insights :

Analysis The dashboard summarizes average customer ratings across key airline service dimensions. These insights help identify strengths, weaknesses, and immediate improvement areas.

Overall Rating – 4.25 (Strong Performance)

- Indicates high overall customer satisfaction.
- Despite weaknesses in specific areas, customers still perceive the airline positively.
- Suggests strong brand trust or good core service delivery.

Cabin Staff Service – 3.01 (Average)

- Service quality is acceptable but inconsistent.
- Customers may be experiencing variations in staff friendliness, responsiveness, or professionalism.
- Training and standardization can improve this score quickly.

Entertainment – 1.44 (Critical Weakness)

- Lowest-rated factor, signaling serious dissatisfaction.
- Possible issues:
  - Outdated or non-functional entertainment systems
  - Limited content options
  - Poor screen quality or connectivity
- Immediate action required, as this heavily impacts long-haul passenger experience.

Food & Beverages – 2.51 (Below Average)

- Customers are not satisfied with meal quality or variety.
- Improvements in menu options, taste, and dietary customization could raise satisfaction.

Ground Service – 2.90 (Moderate Concern)

- Indicates problems during check-in, boarding, or baggage handling.
- Operational delays or staff coordination may be affecting passenger experience.

Seat Comfort – 2.69 (Below Expectations)

- Passengers find seating uncomfortable, especially for long flights.
- Issues may include legroom, seat cushioning, or seat maintenance.

Value for Money – 2.65 (Poor Perception)

- Customers feel the price does not match the service quality.
- Improving low-rated services (entertainment, comfort, food) can directly enhance this perception.

Monthly Trend of Customer Satisfaction

1. **Business Class**
   - Highly volatile satisfaction over the years.
   - Peaks around 2018–2019 and 2022, but frequent sharp drops indicate inconsistent service quality.
   - Suggests issues like variability in cabin service, seat comfort, or value perception.
2. **Economy Class**
   - Shows a clear upward trend after 2020.
   - Reaches highest satisfaction levels by 2022–2023, becoming the most stable segment.
   - Indicates successful improvements in cost-value balance, service standardization, or customer expectations management.
3. **First Class**
   - Strong satisfaction in early years (2017–2018).
   - Continuous decline after 2019, with no strong recovery.
   - Possible reasons: reduced premium services, higher expectations post-COVID, or lack of differentiation from Business Class.

Business Signals

- **Risk Area**: First Class customers – declining loyalty and premium dissatisfaction.
- **Success Area**: Economy Class – service strategies here are working well.
- **Improvement Needed**: Business Class consistency and reliability.

Recommendations

- **First Class**: Reintroduce exclusive services (luxury meals, faster ground services).
- **Business Class**: Standardize service delivery and improve staff training.
- **Economy Class**: Maintain current strategy; use it as a benchmark model.

Aircraft-wise Performance Insights

1. **Top Performers**
    - **Boeing 787-8 →** *Highest customer satisfaction*
        - Known for better cabin pressure, quieter ride, improved air quality.
        - Strongly linked with passenger comfort and long-haul preference.
    - **A320neo**
        - Newer aircraft, fuel-efficient, smoother flights.
        - Positive perception among short- to medium-haul passengers.
2. **Mid-Tier Aircraft**
    - **Boeing 777-200 / 777-200ER**
    - **A330**
        - Stable but not outstanding.
        - Likely affected by aging interiors or inconsistent retrofits.

3. **Low Satisfaction Aircraft (Risk Zone)**
    - **B777-200**
    - **Boeing 747**
    - **A320 (older variant)**
        - Older cabins, higher noise levels, dated in-flight systems.
        - Higher chance of negative reviews and CSAT decline.

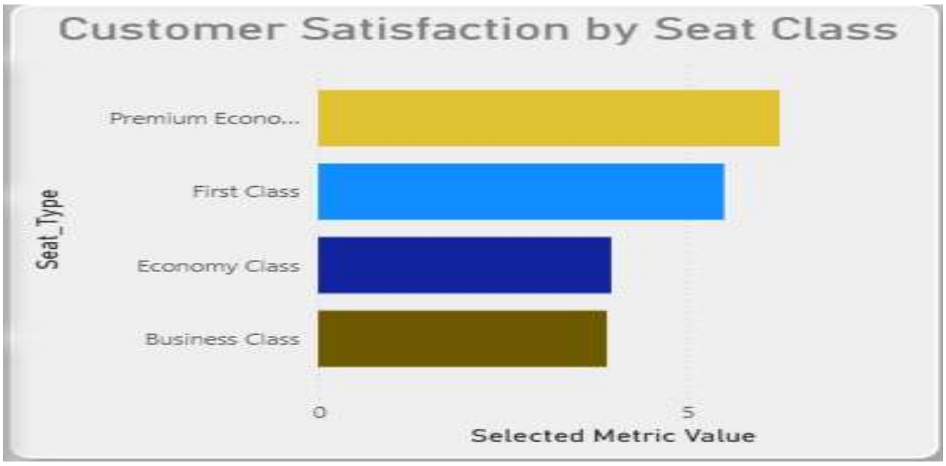
Business Signals

- Customer dissatisfaction is aircraft-driven, not just service-driven.
- Older aircraft correlate strongly with lower satisfaction scores.
- New-generation aircraft directly improve brand perception and loyalty.

Recommendations

- **Prioritize premium & long-haul routes** with:
    - Boeing 787-8, A320neo
- **Retrofit or refurbish**:
    - Boeing 747, older A320, 777-200
- **Route planning strategy**:
    - Deploy high-CSAT aircraft on high-revenue routes.
- **Marketing insight**:
    - Promote “Dreamliner / Neo experience” in branding.

Seat Class Performance Insights



1. **Premium Economy – Highest Satisfaction**
    - Scores highest among all seat classes.
    - Strong balance between price, comfort, and service.
    - Passengers feel they get "more value for money" than other classes.
2. **First Class – High but Not Leading**
    - Satisfaction is high but below Premium Economy.
    - Indicates expectation–experience gap: very high expectations not always met.
3. **Economy Class – Moderate Satisfaction**
    - Acceptable but clearly below premium segments.
    - Satisfaction likely driven by price sensitivity, not comfort.
4. **Business Class – Lowest Satisfaction (Alert Zone)**
    - Surprisingly the lowest-rated class.
    - Signals dissatisfaction in seat comfort, service consistency, or value for money.

Business Signals

- Business Class underperforming despite premium pricing.
- Premium Economy is the strongest competitive advantage.
- First Class needs differentiation to justify premium positioning.

Recommendations

- **Business Class**
    - Improve seat comfort, cabin service standardization, and catering.
    - Re-evaluate pricing vs. perceived value.
- **Premium Economy**
    - Expand availability on more routes.
    - Promote heavily in marketing campaigns.
- **First Class**
    - Enhance exclusivity (personalized service, luxury dining).
- **Economy**
    - Maintain consistency; focus on reliability rather than luxury.

Average Customer Satisfaction by Country

1. **High Satisfaction Regions**
    - Europe & North America
        - Consistently stronger satisfaction levels.
        - Driven by better service standardization, modern aircraft deployment, and mature airport infrastructure.
    - Indicates strong brand trust in developed aviation markets.
2. **Moderate Satisfaction Regions**
    - Asia-Pacific
        - Mixed performance across countries.
        - High growth markets but inconsistent service quality and varying passenger expectations.
3. **Low Satisfaction / Risk Regions**
    - Africa & parts of South America
        - Lower average satisfaction.
        - Likely impacted by ground services, delays, older fleets, and airport facilities.

Business Signals

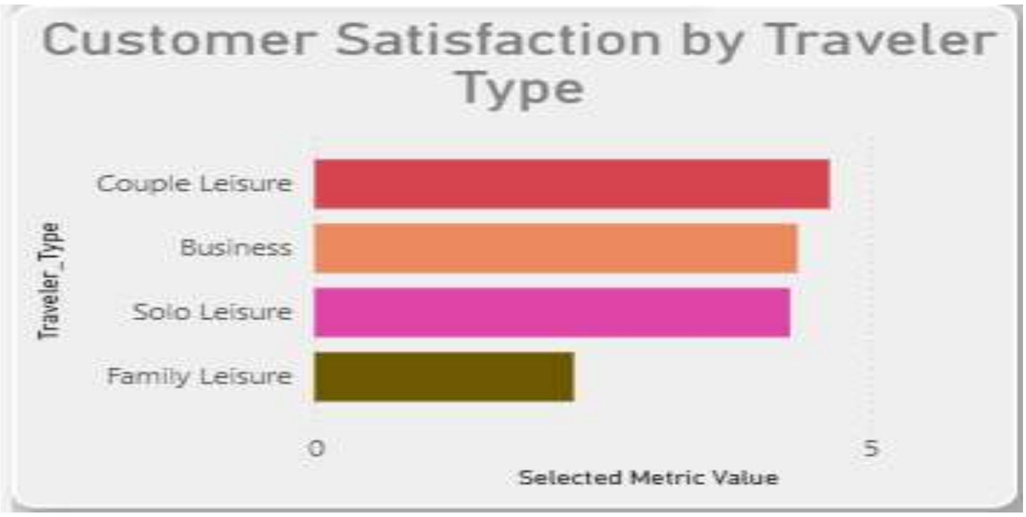
- Satisfaction is region-dependent, not uniform globally.

- Emerging markets show higher variability and risk of negative CSAT.
- Mature markets deliver stable customer experience and loyalty.

Recommendations

- **Europe & North America**
  - Maintain service standards.
  - Use as benchmark regions for global rollout.
- **Asia-Pacific**
  - Targeted improvements in cabin service and punctuality.
  - Localize service offerings to cultural expectations.

Customer Satisfaction by Traveler Type



1. **Couple Leisure – Highest Satisfaction**
   - Top-rated traveler segment.
   - Likely benefits from comfort-focused seating, flexible schedules, and leisure-oriented services.
   - Indicates strong alignment between service offerings and leisure travel expectations.
2. **Business Travelers – High but Sensitive**
   - Satisfaction is high, but slightly below Couple Leisure.

- Highly sensitive to punctuality, lounge access, Wi-Fi, and seat comfort.
- Small disruptions can quickly impact satisfaction.

3. **Solo Leisure – Moderate to High**
    - Good satisfaction, but not the highest.
    - Experience may depend heavily on price, self-service options, and seat availability.
4. **Family Leisure – Lowest Satisfaction**
    - Clearly underperforming.
    - Likely pain points:
        - Seating together
        - Baggage handling
        - Onboard entertainment for children
        - Airport transfer hassles

Business Signals

- Family travelers are the most dissatisfied segment.
- Business travelers represent high-value but high-expectation customers.
- Leisure couples are currently the best-served segment.

Recommendations

- **Family Leisure**
    - Introduce family seating guarantees.
    - Improve kid-friendly meals & entertainment.
- **Business Travelers**
    - Improve on-time performance.
    - Enhance connectivity (Wi-Fi, power outlets).
- **Couple & Solo Leisure**
    - Offer bundled leisure packages.
    - Promote Premium Economy upgrades.

Aircraft & Seat Class Satisfaction Matrix

Aircraft & Seat Class Satisfaction Matrix

Aircraft	Business Class	Economy Class	First Class	Premium Economy	Total
A319	3.50				**3.50**
A320	4.33	5.20			**4.79**
A320neo	8.50				**8.50**
A321	2.00	3.00			**2.50**
A321neo	2.00				**2.00**
A330	7.00				**7.00**
A350				4.33	**4.33**
A380	2.50	3.67			**3.00**
B777-200		6.00			**6.00**
Boeing 737-800		1.00			**1.00**

1. **Top performers**

- **A320neo (Business Class – 8.50)** → Highest satisfaction overall. Indicates strong customer preference for newer aircraft in premium cabins.
- **A330 (Business Class – 7.00)** → Consistently good long-haul comfort perception.
- **B777-200 (Economy – 6.00)** → Best economy experience among wide-body aircraft shown.

2. **Moderate performance**

- **A320 (Business 4.33, Economy 5.20)** → Balanced but not exceptional; acceptable for short–medium haul.
- **A350 (Premium Economy – 4.33)** → Decent but below expectations for a modern aircraft.

3. **Low satisfaction / risk areas**

- **A321 & A321neo (Business ~2.00, Economy ~3.00)** → Clear dissatisfaction, likely due to seat layout, legroom, or service consistency.
- **A380 (Business 2.50, Economy 3.67)** → Surprisingly low for a flagship aircraft → signals aging interiors or service issues.
- **Boeing 737-800 (Economy – 1.00)** → Lowest rated overall → urgent improvement needed.

4. **Class-wise pattern**

- **Business Class scores vary the most** → Aircraft type strongly impacts premium customer satisfaction.
- **Economy Class generally lower and more volatile** → Seat comfort and density are major drivers.

- **Premium Economy appears underutilized** → Few aircraft but mid-level satisfaction.

5. Actionable insights

- Deploy A320neo and A330 on high-value / business-heavy routes.
- Prioritize retrofit or service upgrades for A321 family, A380, and B737-800.
- Improve economy seat comfort & service on narrow-body aircraft.
- Re-evaluate Premium Economy offering on A350 to justify the product positioning.

Distribution of Overall Customer Ratings

1. Strong negative skew

- A large concentration of ratings between 1–2 - indicates many customers are dissatisfied.
- This suggests systemic issues, not isolated incidents.

2. Polarized customer sentiment

- Ratings cluster at very low (1–2) and very high (8–10) values.
- Fewer mid-range ratings (4–6) - customers tend to have extreme experiences (very bad or very good).

3. Limited "average" experiences

- The dip around 5–6 implies fewer neutral journeys.
- Customers are more likely to remember and report poor or excellent service, not average ones.

4. High-rating opportunity

- A noticeable peak around 9–10 shows that when service works well, it works extremely well.
- Best practices from these high-rated journeys should be identified and scaled.

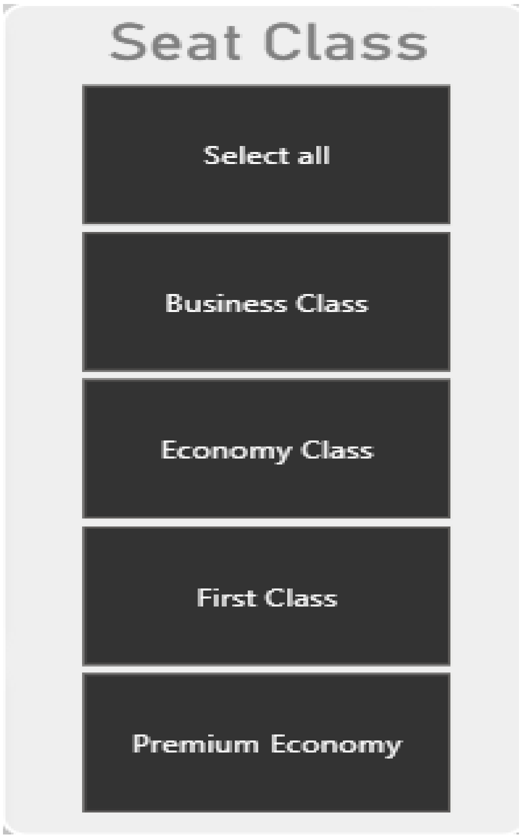
5. Business impact insight

- High volume of low ratings will drag down overall CSAT/NPS, even with some excellent scores.
- Reducing the 1–2 star experiences will have a bigger impact than trying to push 8-10 ratings higher.

6. Immediate actions

- Drill down into 1–2 star reviews by:
    - Aircraft type
    - Seat class
    - Route and delay issues
    - Cabin crew & ground service
- Identify common failure points causing dissatisfaction.
- Replicate service conditions from 9–10 rated flights.

Seat Class Slicer



- **Business Class**
  Shows high sensitivity to *Cabin Staff* and *Seat Comfort*. Small service drops cause sharp rating declines.
- **Economy Class**
  Strongly driven by Value for Money and Seat Comfort. Most low ratings originate here.
- **First Class**
  Ratings depend on personalized service (Cabin Staff + Food). Expectations are highest.

- **Premium Economy**
  Acts as a transition segment—customers compare it with Business Class but pay Economy-like prices.
- **Select All**
  Highlights overall experience imbalance across classes.

Service Metric Slicer

- **Overall Rating**
  Summarizes the end-to-end journey impact.
- **Seat Comfort**
  Largest influencer in **Economy & Premium Economy** dissatisfaction.
- **Cabin Staff**
  Key driver of **high scores (8–10)** across all premium classes.
- **Food**
  Has **low impact in Economy**, but **high impact in Business/First**.
- **Ground Service**
  Often the **hidden reason behind 1–2 ratings**, especially delays & boarding.
- **Value for Money**
  Critical for **Economy customers**—low scores directly lead to negative reviews.

Conclusion

The British Airways Passenger CSAT Dashboard provides valuable real-time insights into customer satisfaction across multiple service dimensions. The analysis reveals that while the overall customer rating is high, several critical service areas such as in-flight entertainment, seat comfort, and value for money require significant improvement. The dashboard highlights clear variations in satisfaction based on aircraft type, seat class, and geographic regions. Newer aircraft models and premium seat classes demonstrate higher customer satisfaction compared to older aircraft and economy class. The use of interactive visualizations enables data-driven decision-making and performance monitoring. Overall, this project demonstrates how business intelligence tools like Power BI can effectively transform customer feedback into actionable insights for service enhancement.